Git Merge

Introduction

Imagine you are working on a **software project** with a team. Each developer is working on a **separate feature branch**. Once a feature is complete, it needs to be **merged into the main project**.

How do you combine all changes without losing data?

This is where the git merge command helps!

Section 1: Learn

What is git merge?

git merge **is a Git command used to combine changes from one branch into another.** It allows developers to:

1. **Merge feature branches into the main branch**.
2. **Integrate multiple changes efficiently**.
3. **Maintain a clean and organized project history**.

Why Do We Need git merge?

Without git merge	**With** git merge
Changes stay isolated in branches	Combines work into the main branch
No way to integrate new features	Adds new features to the project
Developers overwrite each other's work	Maintains a clean and structured history

Without git merge	**With** git merge
Hard to track project progress	All updates are merged and tracked

How Does git merge Work?

1. **Switch to the main branch** where you want to merge changes.
2. **Run** git merge branch-name to merge changes from another branch.
3. **Resolve conflicts if any** (when two branches change the same file).
4. **Commit the merge** to finalize the changes.

An Interesting Anecdote

Before Git, developers manually copied and pasted code changes, leading to **errors and lost work**. With **Git merge**, teams can collaborate **smoothly and efficiently**, ensuring **all contributions are integrated properly**.

Section 2: Practice

Step-by-Step Guide to Using git merge

1. Installing Git (If Not Installed)

```
# Check if Git is installed
git --version

# If Git is not installed, install it (Ubuntu)
sudo apt update
sudo apt install git
```

Why is this important?

- Ensures Git is **installed and ready to use**.
- Allows you to **merge branches and track changes**.

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2. Creating and Merging a Feature Branch

```
# Create a new branch for a feature
git branch feature-1

# Switch to the new branch
git checkout feature-1

# Create a new file and add some code
echo "print('Feature 1')" > feature.py

# Add and commit the file
git add feature.py
git commit -m "Added feature 1"

# Switch back to the main branch
git checkout main

# Merge the feature branch into main
git merge feature-1
```

What does this do?

- Creates a new branch (feature-1) and adds a file.
- Merges feature-1 into main without losing any data.

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3. Checking the Merge History

```
# View the commit history
git log --oneline --graph
```

Why is this useful?

- Shows how branches were merged into the project.
- Helps in **tracking past merges**.

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4. Handling Merge Conflicts

Sometimes, merging causes **conflicts** if the same file is modified in two branches.

```
# Show conflicting files
git status

# Open the conflicting file and manually fix the issue

# After resolving, stage the file
git add conflicted-file.txt

# Commit the resolved merge
git commit -m "Resolved merge conflict"
```

How does this help?

- Prevents accidental overwriting of changes.
- Ensures a **smooth integration** of work.

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5. Merging Without a Merge Commit (Fast-Forward Merge)

If no changes were made in the main branch, Git performs a **fast-forward merge**.

```
git checkout main
git merge feature-1 --ff-only
```

What does this do?

- Moves the main branch pointer to the latest commit from feature-1.
- Avoids extra merge commits, keeping the history **clean**.

Real-World Example

A **tech startup** is developing a **shopping app**. Their **team of developers** is working on different features:

- **Developer A** is building the **cart system**.
- **Developer B** is working on **payment integration**.
- **Developer C** is designing the **checkout process**.

Without Merging:

- Changes stay in separate branches and **never reach the main project**.
- Developers **cannot see each other's work**, causing integration delays.

With Merging (git merge**)**:

1. Each feature is developed in **separate branches**.
2. Once tested, **branches are merged into the main project**.
3. The app **grows step by step without conflicts**.

Result? Faster development, **clean code, and efficient teamwork**.

Section 3: Know More

Frequently Asked Questions

1. **What is the difference between** git merge **and** git rebase**?**
    - git merge **combines branches and keeps history**.
    - git rebase **replays commits from one branch onto another**.
2. **Can I undo a merge?**

```
git merge --abort  # If the merge is not committed yet


git reset --hard HEAD~1  # Undo a completed merge
```

3. This **restores the previous state** before merging.
4. **How do I check which branches are already merged?**

```
git branch --merged
```

5. This lists **all branches that have been merged** into the current branch.
6. **What is a "merge commit"?**

   A commit created when merging branches that **records the merge operation**.
7. **Where can I practice Git merging?**
    - Create a **GitHub repository** and test merging different branches.
    - Work on **collaborative projects** to understand real-world merging scenarios.

Conclusion

The git merge command is **essential for combining different branches and integrating new features**. It enables developers to **work independently and merge changes efficiently**.

Start by **creating branches**, experiment with **merging them**, and soon you'll be **managing team projects smoothly!**